



Model (B)

رقم الكشف:

الفصل:

الإسم:

(Put (T) for true and (F) for false)

(1) Fleming's left hand rule used to determine the direction of magnetic torque acting on a coil carrying a current and putted in magnetic field. ☐

(2) If the current in the coil is doubled the magnetic torque acted on it will halved. ☐

(3) The magnetic flux density unit is equivalent to (N / A . m). ☐

(4) If the magnetic flux increases four times the magnetic torque also increases four times. ☐

(MCQ)

(5) A rectangular loop (7×10^{-4}) of 12 turns, carrying current of 3A, is placed in magnetic field of 0.53 T flux density such that the plane of the loop is parallel the field the magnetic dipole moment equal A . m^2

(a) Zero

(b) 0.0252

(c) 252

(d) 0.252

(6) A straight wire 5 m long through which passes a current 3 A. The wire is placed perpendicular to a magnetic field of magnetic flux density .2 T. Then the force acting on the wire

(a) 30 N

(b) 300 N

(c) 0.3 N

(d) 0.03 N

(Put (T) for true and (F) for false)

(7) Emf induces must produces I induces. ☐

(8) Emf induces is direct proportional to the magnetic flux. ☐

(9) Emf induces must be in a direction such as to help the change producing it. ☐

(10) Increasing the current intensity in primary coil by using rheostat produces Backward induces current. ☐

(11) Switching off the current in primary coil produces forward induces current
In the secondary coil. ☐

(12) The scientific base of electric transformer is the mutual induction. ☐

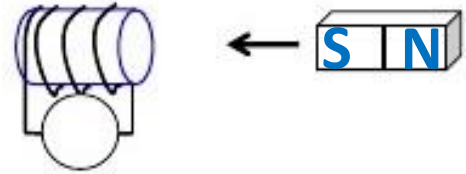
(MCQ)

(13) In faraday Experiment: the Emf induced is direct proportional to

- (a) The strength of the magnet
- (b) The rate of change of magnetic flux
- (c) The magnetic flux density
- (d) the length of the coil

(14) In the opposite figure: what is the pole formed in the area faced the magnet

- a) South pole
b) North pole
d) we can't determine



(15) In faraday Experiment: the direction of induced current is generated in the conductor depends on

- (a) The direction of the motion only (b) The direction of the magnetic field only
- (c) The direction of the magnetic field and the motion (d) None of the above

(16) the equivalent unit for (Weber)

- (a) Volt. sec (b) Volt / sec (c) Tesla . m (d) Tesla / m^2

(17) An induced EMF of value 15 V is generated in a coil whose number of turns 333 turns , when the rate of change of the magnetic flux through it is

- (a) - 0.045 Wb/sec (b) + 0.045 Wb/sec (c) + 0.015 Wb/sec (d) - 0.01 Wb/sec

(18) Forward current is in the same direction of

- (a) Induces current in primary coil
- (b) Original current in primary coil
- (c) Induces current in secondary coil
- (d) Original current in secondary coil

(19) An inductance of 0.02 H exist between two coils if a current of 3A is decay in 0.001 sec what is value of the EMF induced

- (a) 60 volt (b) 6 volt (c) 120 volt (d) 30 volt

(20) Henry is equivalent to

- (a) Volt . sec (b) A . sec (c) Volt . Sec / A (d) A . Sec / volt